```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('Telco_Customer_Churn.csv')

df
```

```
      customerID  gender  SeniorCitizen Partner Dependents  tenure  \
0     7590-VHVEG  Female              0     Yes         No       1
1     5575-GNVDE    Male              0      No         No      34
2     3668-QPYBK    Male              0      No         No       2
3     7795-CFOCW    Male              0      No         No      45
4     9237-HQITU  Female              0      No         No       2
...          ...     ...            ...     ...        ...     ...
7038  6840-RESVB    Male              0     Yes        Yes      24
7039  2234-XADUH  Female              0     Yes        Yes      72
7040  4801-JZAZL  Female              0     Yes        Yes      11
7041  8361-LTMKD    Male              1     Yes         No       4
7042  3186-AJIEK    Male              0      No         No      66

     PhoneService     MultipleLines InternetService
OnlineSecurity  ...  \
0              No  No phone service             DSL
No  ...
1             Yes                No             DSL
Yes  ...
2             Yes                No             DSL
Yes  ...
3              No  No phone service             DSL
Yes  ...
4             Yes                No     Fiber optic
No  ...
...           ...               ...             ...             ...  ..
.
7038          Yes               Yes             DSL
Yes  ...
7039          Yes               Yes     Fiber optic
No  ...
7040           No  No phone service             DSL
Yes  ...
7041          Yes               Yes     Fiber optic
No  ...
7042          Yes                No     Fiber optic
Yes  ...

     DeviceProtection TechSupport StreamingTV StreamingMovies
Contract  \
0                  No          No          No              No  Month-
```

```
to-month
1                  Yes          No          No              No
One year
2                   No          No          No              No  Month-
to-month
3                  Yes         Yes          No              No
One year
4                   No          No          No              No  Month-
to-month
...                ...         ...         ...             ...
...
7038               Yes         Yes         Yes             Yes
One year
7039               Yes          No         Yes             Yes
One year
7040                No          No          No              No  Month-
to-month
7041                No          No          No              No  Month-
to-month
7042               Yes         Yes         Yes             Yes
Two year

     PaperlessBilling              PaymentMethod MonthlyCharges
TotalCharges  \
0                  Yes           Electronic check          29.85
29.85
1                   No               Mailed check          56.95
1889.5
2                  Yes               Mailed check          53.85
108.15
3                   No  Bank transfer (automatic)          42.30
1840.75
4                  Yes           Electronic check          70.70
151.65
...                ...                        ...            ...
...
7038               Yes               Mailed check          84.80
1990.5
7039               Yes    Credit card (automatic)         103.20
7362.9
7040               Yes           Electronic check          29.60
346.45
7041               Yes               Mailed check          74.40
306.6
7042               Yes  Bank transfer (automatic)         105.65
6844.5

     Churn
0       No
```

```
1       No
2      Yes
3       No
4      Yes
...    ...
7038    No
7039    No
7040    No
7041   Yes
7042    No

[7043 rows x 21 columns]
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

REPLCAED TENURE AS 0 , AND NO. OF TOTAL CHRGEES ARE RECORDED

```
df["TotalCharges"] = df["TotalCharges"].replace(" ","0")
df["TotalCharges"] = df["TotalCharges"].astype("float")

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   float64
 20  Churn             7043 non-null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1+ MB
```

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

```
df.describe()
```

```
       SeniorCitizen       tenure  MonthlyCharges  TotalCharges
count    7043.000000  7043.000000     7043.000000   7043.000000
mean        0.162147    32.371149       64.761692   2279.734304
std         0.368612    24.559481       30.090047   2266.794470
min         0.000000     0.000000       18.250000      0.000000
25%         0.000000     9.000000       35.500000    398.550000
```

```
50%          0.000000    29.000000       70.350000   1394.550000
75%          0.000000    55.000000       89.850000   3786.600000
max          1.000000    72.000000      118.750000   8684.800000
```

```
df["customerID"].duplicated().sum()
```

```
np.int64(0)
```

CONVERTED 0 AND 1 VALUES "SENIOR CITIZEN"

```
def conv(value):
    if value ==1:
        return "yes"
    else :
        return "no"
df['SeniorCitizen'] = df["SeniorCitizen"].apply(conv)
```

```
df.head(20)
```

```
    customerID  gender SeniorCitizen Partner Dependents  tenure
PhoneService  \
0   7590-VHVEG  Female            no     Yes         No       1
No
1   5575-GNVDE    Male            no      No         No      34
Yes
2   3668-QPYBK    Male            no      No         No       2
Yes
3   7795-CFOCW    Male            no      No         No      45
No
4   9237-HQITU  Female            no      No         No       2
Yes
5   9305-CDSKC  Female            no      No         No       8
Yes
6   1452-KIOVK    Male            no      No        Yes      22
Yes
7   6713-OKOMC  Female            no      No         No      10
No
8   7892-POOKP  Female            no     Yes         No      28
Yes
9   6388-TABGU    Male            no      No        Yes      62
Yes
10  9763-GRSKD    Male            no     Yes        Yes      13
Yes
11  7469-LKBCI    Male            no      No         No      16
Yes
12  8091-TTVAX    Male            no     Yes         No      58
```

```
Yes
13  0280-XJGEX    Male            no      No         No      49
Yes
14  5129-JLPIS    Male            no      No         No      25
Yes
15  3655-SNQYZ  Female            no     Yes        Yes      69
Yes
16  8191-XWSZG  Female            no      No         No      52
Yes
17  9959-WOFKT    Male            no      No        Yes      71
Yes
18  4190-MFLUW  Female            no     Yes        Yes      10
Yes
19  4183-MYFRB  Female            no      No         No      21
Yes

       MultipleLines InternetService       OnlineSecurity  ...  \
0   No phone service             DSL                   No  ...
1                 No             DSL                  Yes  ...
2                 No             DSL                  Yes  ...
3   No phone service             DSL                  Yes  ...
4                 No     Fiber optic                   No  ...
5                Yes     Fiber optic                   No  ...
6                Yes     Fiber optic                   No  ...
7   No phone service             DSL                  Yes  ...
8                Yes     Fiber optic                   No  ...
9                 No             DSL                  Yes  ...
10                No             DSL                  Yes  ...
11                No              No  No internet service  ...
12               Yes     Fiber optic                   No  ...
13               Yes     Fiber optic                   No  ...
14                No     Fiber optic                  Yes  ...
15               Yes     Fiber optic                  Yes  ...
16                No              No  No internet service  ...
17               Yes     Fiber optic                  Yes  ...
18                No             DSL                   No  ...
19                No     Fiber optic                   No  ...

       DeviceProtection          TechSupport          StreamingTV  \
0                    No                   No                   No
1                   Yes                   No                   No
2                    No                   No                   No
3                   Yes                  Yes                   No
4                    No                   No                   No
5                   Yes                   No                  Yes
6                    No                   No                  Yes
7                    No                   No                   No
8                   Yes                  Yes                  Yes
9                    No                   No                   No
```

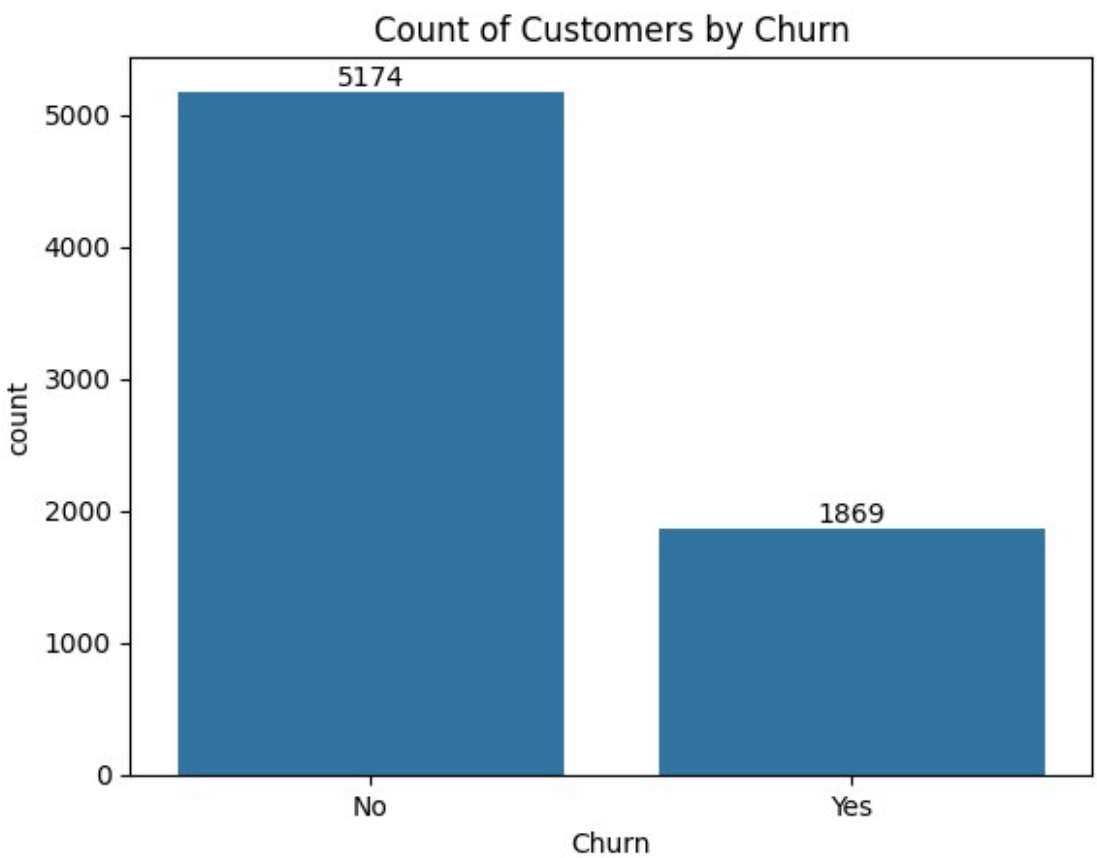
10	No	No	No
11	No internet service	No internet service	No internet service
12	Yes	No	Yes
13	Yes	No	Yes
14	Yes	Yes	Yes
15	Yes	Yes	Yes
16	No internet service	No internet service	No internet service
17	Yes	No	Yes
18	Yes	Yes	No
19	Yes	No	No

	StreamingMovies	Contract	PaperlessBilling \
0	No	Month-to-month	Yes
1	No	One year	No
2	No	Month-to-month	Yes
3	No	One year	No
4	No	Month-to-month	Yes
5	Yes	Month-to-month	Yes
6	No	Month-to-month	Yes
7	No	Month-to-month	No
8	Yes	Month-to-month	Yes
9	No	One year	No
10	No	Month-to-month	Yes
11	No internet service	Two year	No
12	Yes	One year	No
13	Yes	Month-to-month	Yes
14	Yes	Month-to-month	Yes
15	Yes	Two year	No
16	No internet service	One year	No
17	Yes	Two year	No
18	No	Month-to-month	No
19	Yes	Month-to-month	Yes

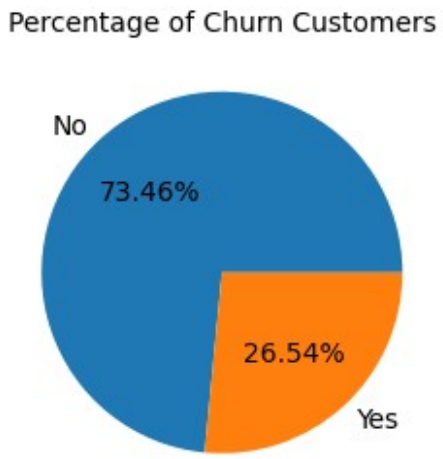
	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.50	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes
5	Electronic check	99.65	820.50	Yes
6	Credit card (automatic)	89.10	1949.40	No
7	Mailed check	29.75	301.90	No
8	Electronic check	104.80	3046.05	Yes
9	Bank transfer (automatic)	56.15	3487.95	No
10	Mailed check	49.95	587.45	No
11	Credit card (automatic)	18.95	326.80	No
12	Credit card (automatic)	100.35	5681.10	No
13	Bank transfer (automatic)	103.70	5036.30	Yes
14	Electronic check	105.50	2686.05	No

```
15    Credit card (automatic)         113.25       7895.15     No
16               Mailed check          20.65       1022.95     No
17  Bank transfer (automatic)         106.70       7382.25     No
18    Credit card (automatic)          55.20        528.35    Yes
19           Electronic check          90.05       1862.90     No

[20 rows x 21 columns]
```

```
ax = sns.countplot(x = 'Churn', data = df)
ax.bar_label(ax.containers[0])
plt.title ("Count of Customers by Churn")
plt.show()
```



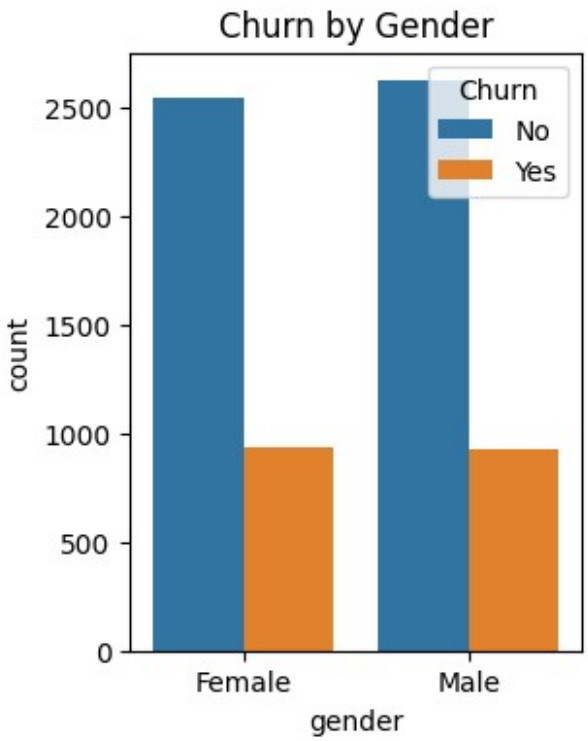

```
plt.figure(figsize = (3,4))
gb = df.groupby("Churn").agg({'Churn':"count"})
plt.pie(gb['Churn'],labels = gb.index, autopct = "%1.2f%%")
plt.title ("Percentage of Churn Customers", fontsize = 10)
plt.show()
```



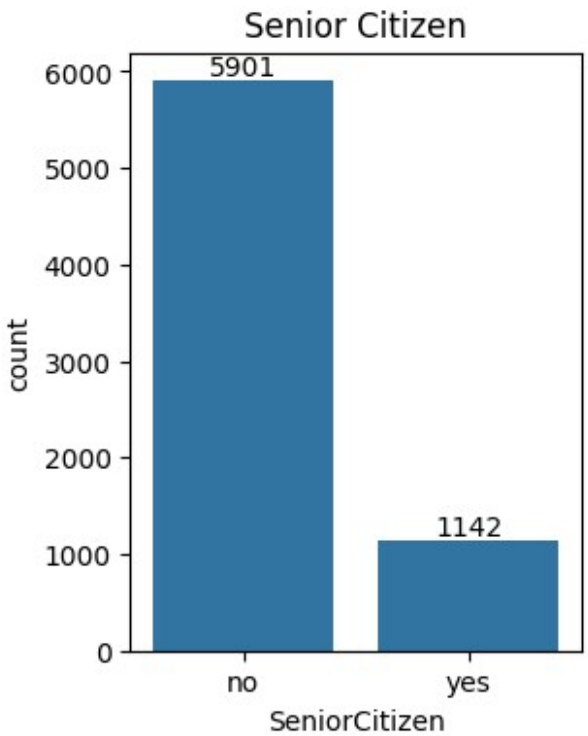
FROM THE GIVEN PIE CHART WE CAN CONCLUDE THAT 26.54% OF OUR CUSTOMERS HAVE CHURNED OUT

NOW LETS EXPLORE THE REASON BEHIND IT.

```
plt.figure(figsize = (3,4))
sns.countplot (x ="gender", data = df , hue = 'Churn')
plt.title("Churn by Gender")
plt.show()
```



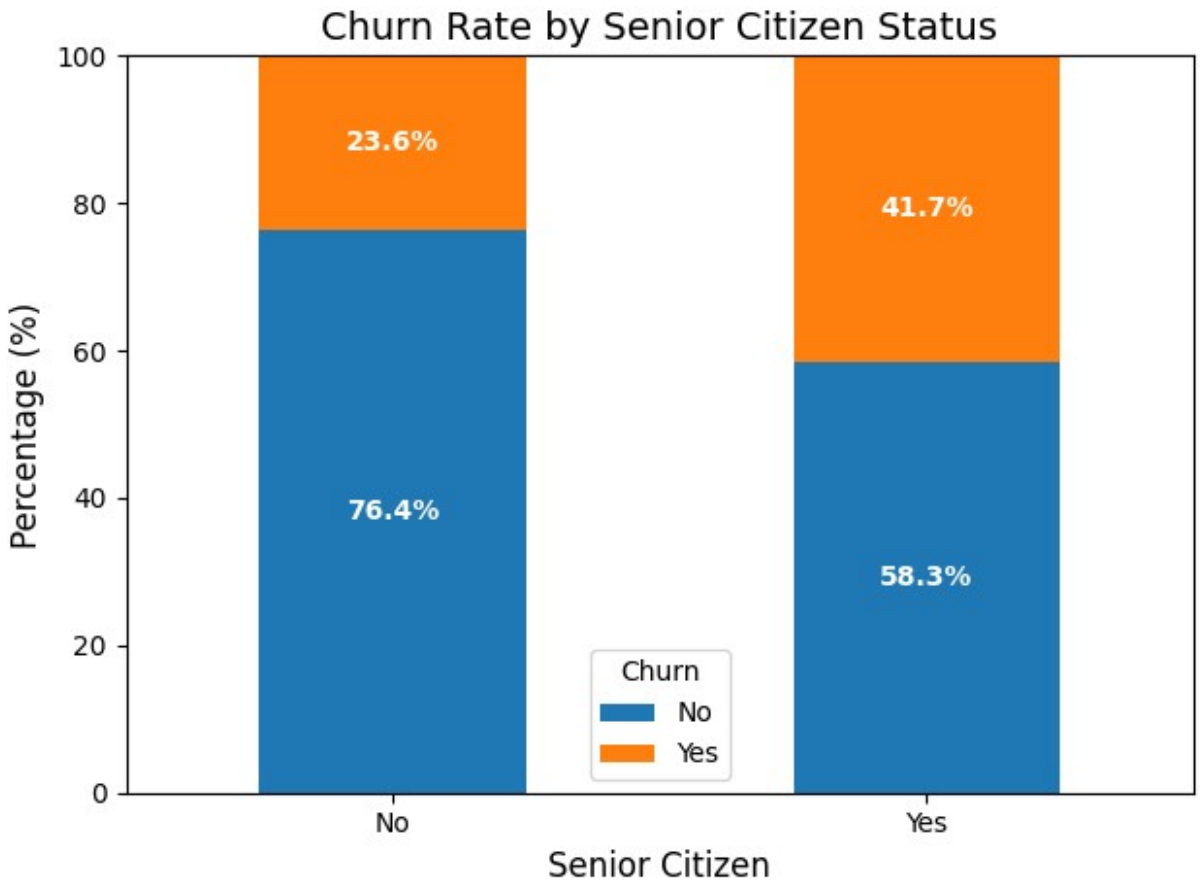

```
plt.figure(figsize = (3,4))
ax = sns.countplot (x ="SeniorCitizen", data = df )
ax.bar_label(ax.containers[0])
plt.title("Senior Citizen")
plt.show()
```




```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create a crosstab to get percentages
ct = pd.crosstab(df['SeniorCitizen'], df['Churn'])
ct_pct = ct.apply(lambda x: x/x.sum() * 100, axis=1)

# Create the stacked bar chart
plt.figure(figsize=(8, 6))
ct_pct.plot(kind='bar', stacked=True)

# Add percentage labels on each segment
for i, (name, row) in enumerate(ct_pct.iterrows()):
    prev = 0
    for j, val in enumerate(row):
        plt.text(i, prev + val/2, f'{val:.1f}%',
                 ha='center', va='center',
                 fontweight='bold', color='white')
        prev += val

# Customize the plot
plt.title("Churn Rate by Senior Citizen Status", fontsize=14)
plt.xlabel("Senior Citizen", fontsize=12)
plt.ylabel("Percentage (%)", fontsize=12)
plt.xticks([0, 1], ['No', 'Yes'], rotation=0)
plt.legend(title='Churn')
```

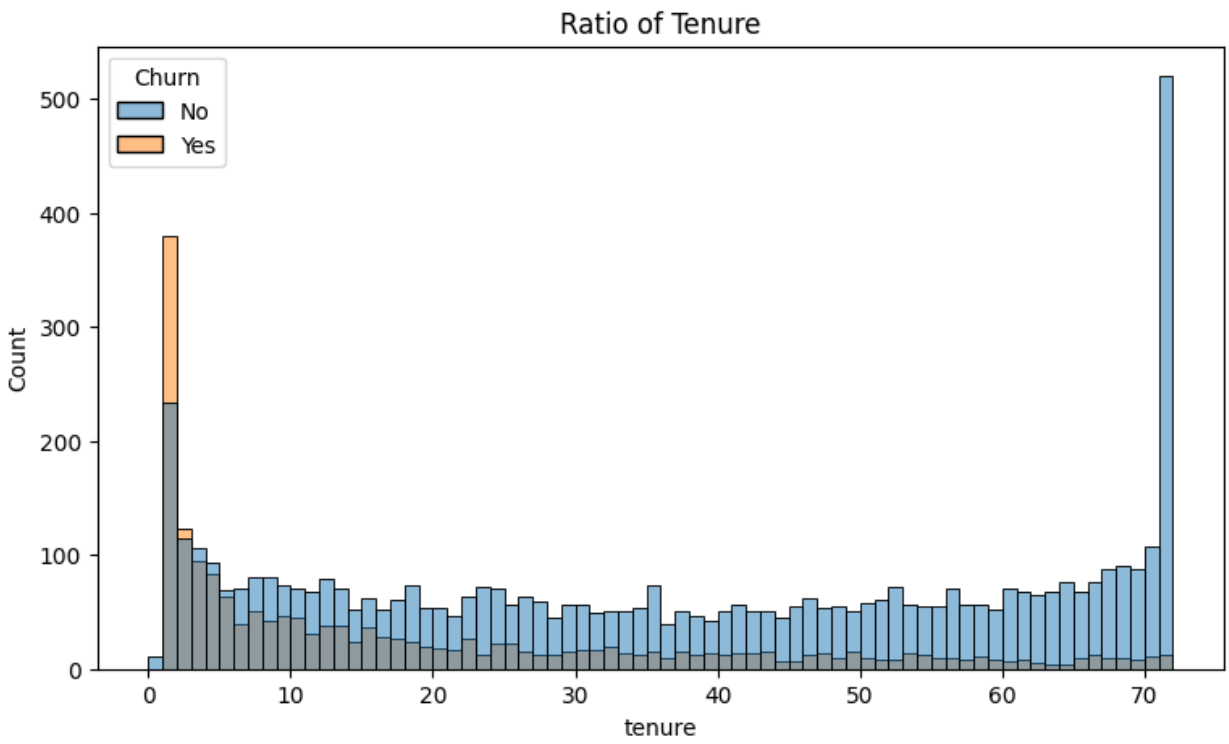
```
plt.ylim(0, 100)

plt.tight_layout()
plt.show()
```

```
<Figure size 800x600 with 0 Axes>
```



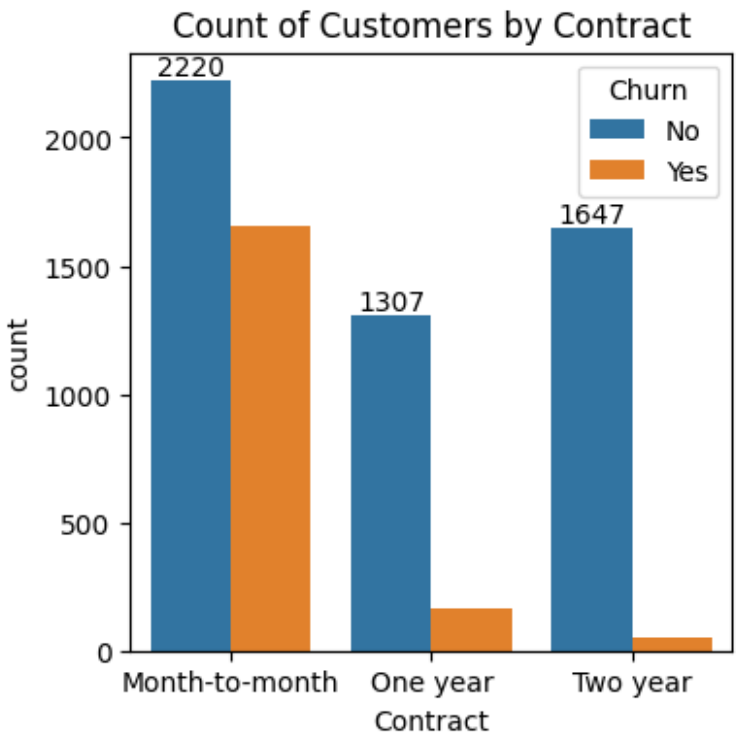
COMPARATIVELY A GREATER PERCENTAGE OF PEOPLE IN SENIOR CITIZEN CATEGORY HAVE CHURNED OUT

```
plt.figure(figsize = (9,5))
sns.histplot (x ="tenure", data = df, bins = 72 , hue = 'Churn')
plt.title("Ratio of Tenure")
plt.show()
```



PEOPLE WHO HAVE USED OUR SERVICES FOR A LONG TIME HAVE STAYED AND PEOPLE WHO HAVE USED OUR SERVUCES FOR 1 OR 2 MONTHS HAVE CHURNED

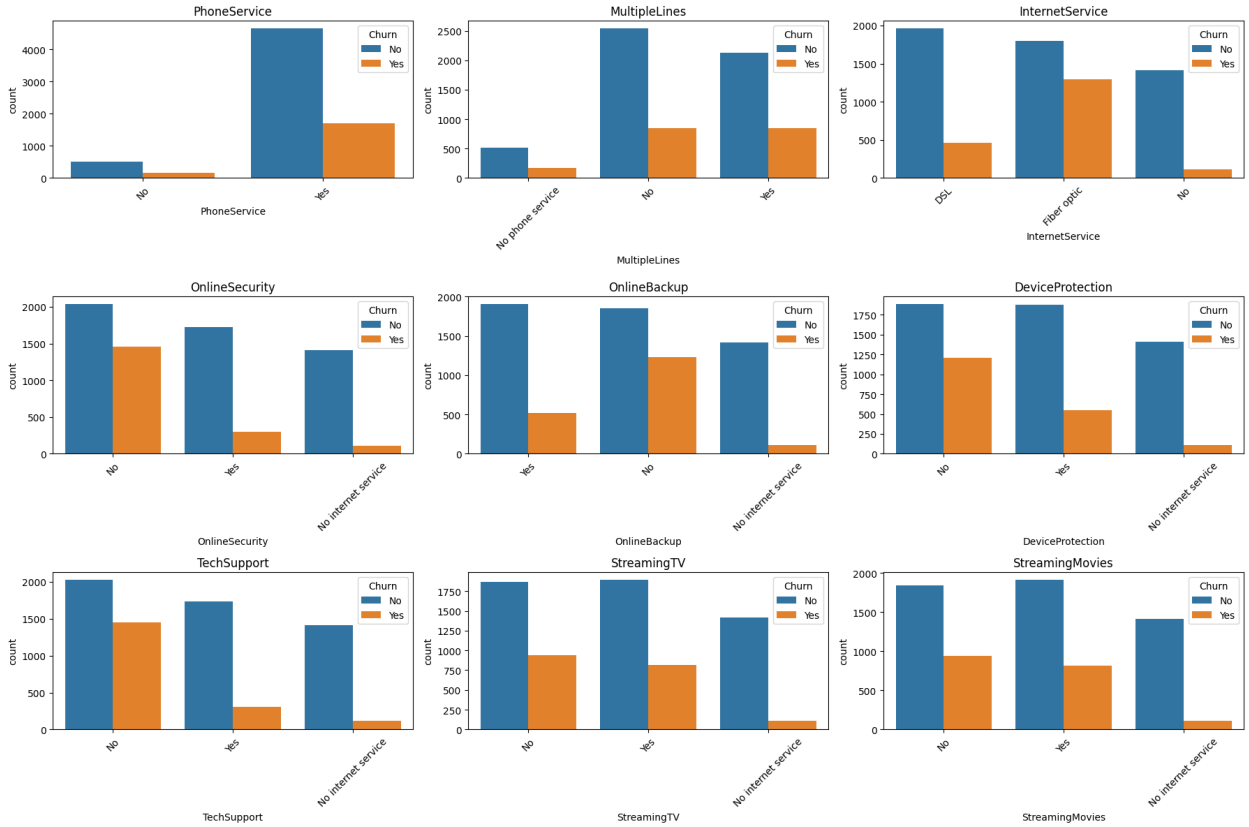
```
plt.figure(figsize = (4,4))
ax = sns.countplot(x = 'Contract', data = df , hue = 'Churn')
ax.bar_label(ax.containers[0])
plt.title ("Count of Customers by Contract")
plt.show()
```



PEOPLE WHO HAVE MONTH_TO_MONTH CONTRACT ARE LIKELY TO CHURN THEN THE YEAR WISE CONTRACTS

```
df.columns.values

array(['customerID', 'gender', 'SeniorCitizen', 'Partner',
'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
       'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',
       'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',
       'TotalCharges', 'Churn'], dtype=object)



cols = [
'PhoneService', 'MultipleLines', 'InternetService',
'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
'TechSupport', 'StreamingTV', 'StreamingMovies'
]

plt.figure(figsize=(18,12))
```

```
for i, col in enumerate(cols, 1):
    plt.subplot(3, 3, i)
    sns.countplot(data=df, x=col, hue = df["Churn"])
    plt.title(col)
    plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```



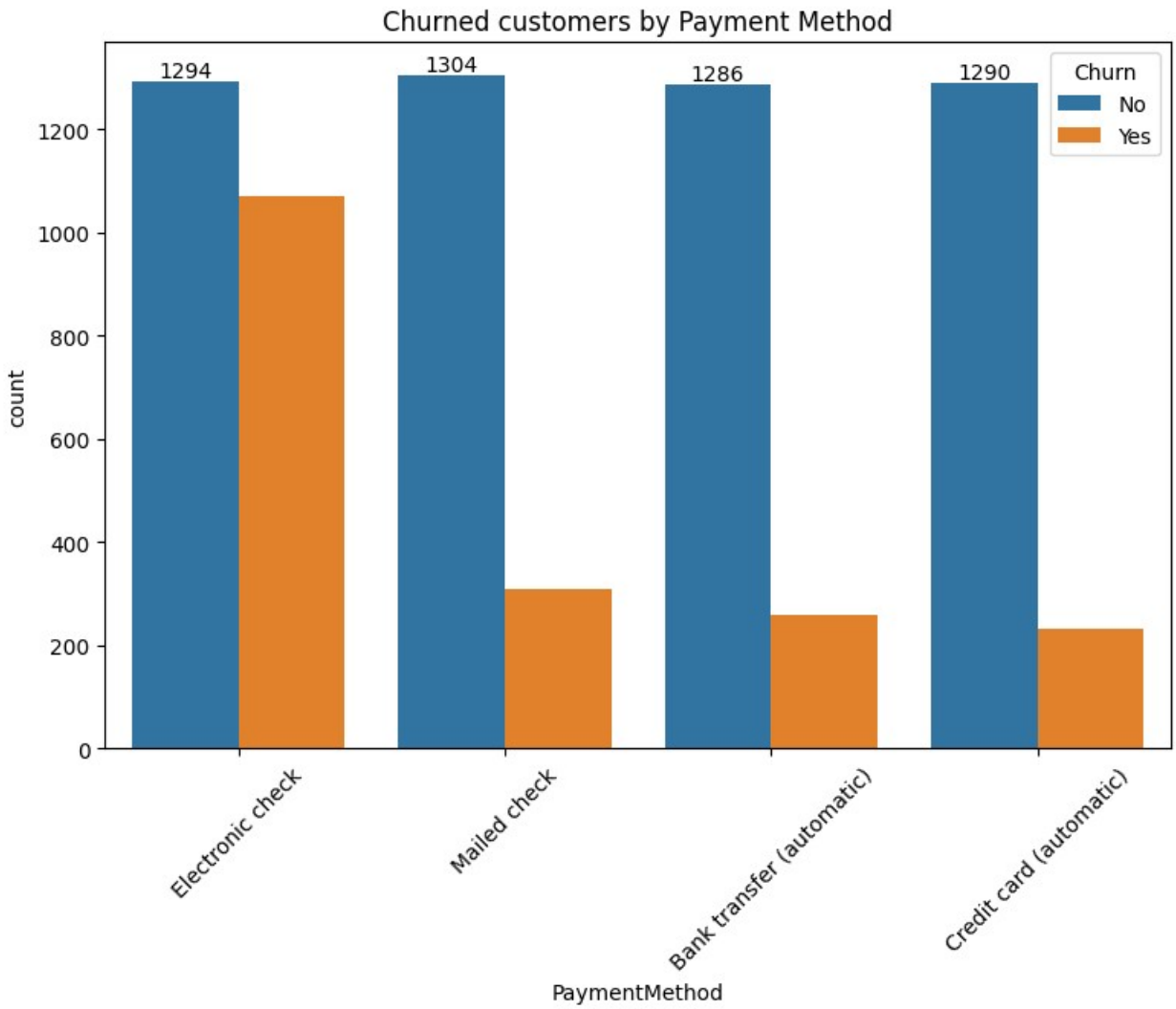
#Customers who do not have OnlineSecurity, TechSupport, or DeviceProtection show significantly higher churn, indicating these services help retain customers.

Fiber optic InternetService users have the highest churn rate compared to DSL or customers without internet service.

Customers with StreamingTV and StreamingMovies do not show a strong reduction in churn, suggesting entertainment services alone do not drive retention.

Overall, value-added support and security services appear to reduce churn more effectively than basic connectivity or streaming services.

```
plt.figure(figsize = (9,6))
ax = sns.countplot(x = 'PaymentMethod', data = df, hue = "Churn")
ax.bar_label(ax.containers[0])
plt.title ("Churned customers by Payment Method")
plt.xticks(rotation = 45)
plt.show()
```



CUSTOMER IS LIKELY TO CHECK WHEN USING ELECTRONIC CHECK